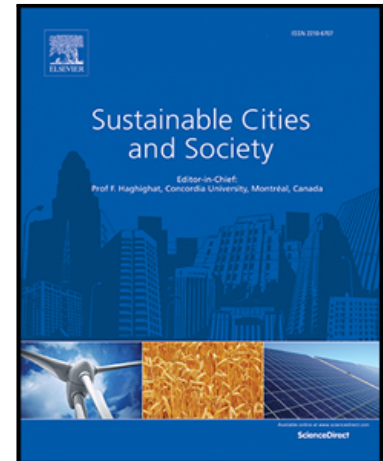


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Highlights

- Hourly temperature variability (HTV) is associated with increased mental health risks
- The highest cumulative effects occur over lag 0–1 day
- Sex and age differences were observed in association between HTV and mental health
- Winter shows delayed cumulative impacts extending to lag 0–12 days

Subtropical Climate and Intraday Temperature Variability: Mental Health Risks and Modifying Roles of Demographic and Seasonal Factors in Texas

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Abstract

Objectives. To examine the relationships of intra-day hourly temperature variability (HTV) with mental health outcomes in subtropical climates and effect modification by demographic characteristics and season.

Methods. We applied a time-series meta-analytic framework to analyze daily temperature conditions and hospital visits across the 53 largest Texas counties between 2015 and 2019. A generalized additive model combined with a distributed lag non-linear model was used to examine exposure-lag-response

relationships in each region, and a random effects meta-analysis was applied to estimate the pooled effects across counties, followed by stratified analyses to assess effect modification.

Results. The highest cumulative mental health risks were observed over lag 0-1 days, with increased intra-day HTV associated with higher risks of mental health-related outpatient hospital visits. Compared to young adults, children/youths were at greater risk of outpatient hospital visits for overall mental and behavioral disorders (F00-F99) associated with HTV, whereas older adults showed greater increases in outpatient visits for substance use disorders (F10-F19) in relation to HTV. Additionally, effects peaked at lag 0-1 in most seasons, but winter showed delayed and prolonged cumulative effects.

Conclusions. The seasonal differences underscore the need to pay particular attention to the heightened risks in spring and prolonged risks in winter. Additionally, greater susceptibility of children/youth and older adults to HTV-related mental disorders suggests that healthcare systems and school-based mental health services should incorporate seasonal monitoring protocols and enhance mental health screening and outreach during high-risk periods of elevated temperature variability in subtropical regions.

Keywords

Hourly temperature variability, Mental health, Subtropical climate, Time-series analysis.

1. Introduction

Mental diseases are among the leading contributors to global disease burden, affecting an estimated 970 million people (12% of the global population).¹ Research on the relationship of mental and behavioral disorders (MBDs) to temperature exposure is rapidly growing, with particular emphasis on the disproportionate impacts observed among vulnerable groups and across regional contexts.^{2,3}

Epidemiological research has largely focused on the mental health impact of high temperatures and extreme events, with most studies relying on absolute metrics (e.g., daily mean, minimum, and maximum temperatures) and reporting associations with higher likelihood of mental health conditions such as suicide and neurotic disorders.⁴⁻⁶ Extreme events have also been frequently linked to elevated rates of

hospital admission for MBDs.^{3,4,7} However, temperature variability (TV) measures (e.g., diurnal temperature range, standard deviation of daily temperature records), which reflect short-term temperature fluctuations, have been examined far less frequently in mental health studies. A recent systematic review on the health impact of TV noted that evidence on its relationship to mental health remains limited, with most studies focusing on schizophrenia and being geographically concentrated in China.⁸ Consistent evidence indicates an association of short-term exposure to TV with higher risk for schizophrenia;⁹⁻¹¹ however, associations of TV with other mental diseases, such as mood or anxiety disorders, have been examined in only a few studies.^{12,13} This paucity of research limits our understanding of the broader mental health impacts of TV and their generalization beyond China; as such, further research is required to address a broader spectrum of MBDs across other global contexts and climate zones.

Texas is the largest state by area in the contiguous United States, the second most populous state in the nation, and is largely located within the subtropical climate zone. Mental diseases pose a significant public health challenge in Texas, with widespread disparities in access to healthcare services.¹⁴ Notably, in warm subtropical climates like Texas, short-term temperature variability may impose greater heat stress than in cooler climate zones. Subtropical regions are often characterized by persistently high temperatures, which have been linked to increased incidence of heat-related illnesses.¹⁵ Under such conditions, rapid temperature fluctuations may exceed physiological heat-tolerance thresholds, compromise thermoregulation capacity, and thereby intensify health risks.¹⁶ In addition, compound exposure to high temperature and elevated humidity can reduce the efficiency of the human body's natural evaporative cooling,¹⁷ amplifying the perceived thermal load of daily variability relative to drier and cooler climates and prolonging period of physiologically dangerous conditions.¹⁸ Although populations in warm regions are generally acclimatized to persistently high temperatures, abrupt TV can disrupt this adaptation and induce greater thermal strain; meanwhile, populations in cooler climates are more acclimatized to broader seasonal variation and thus exhibit greater physiological flexibility and resilience to temperature fluctuation.¹⁹ Prior research on the health impact of temperature in Texas has

revealed that daily temperature characteristics are closely linked to risks in all-cause emergency department (ED) visits and mortality.²⁰⁻²² Likewise, studies in other subtropical regions, such as Hong Kong and Florida, have reported that high temperatures are significantly associated with greater risks of ED visits, hospitalization, and mortality.²³⁻²⁵ Consistent evidence of a positive relationship underscores the importance of considering daily temperature exposure as a critical determinate of public health in subtropical settings.

However, existing research on the health impact of short-term TV exposure has largely been conducted in temperate climates. For example, strong evidence linking increased daily TV to elevated mortality risks has been reported in United Kingdom^{26,27} and Japan.^{28,29} Similarly, studies conducted in temperate regions of China have revealed significant associations between short-term TV and hospital admissions for cardiovascular and cerebrovascular diseases,³⁰⁻³⁴ as well as mental disorders.^{9,10,13} Association of TV and mental disorders has also been found in New York state, United States.¹² In contrast, evidence from subtropical climates remains scarce, with only a limited number of studies examining the associations between short-term TV and mortality and cardiovascular and respiratory disease outcomes. For example, Luo et al.³⁵ reported that diurnal temperature range (DTR) was significantly associated with increased non-accidental mortality in a subtropical megacity in China. Li et al.³⁶ found that increases in DTR were associated with higher emergency room admissions for respiratory tract infection. Moreover, significant associations of DTR with cardiovascular disease-related morbidity and mortality were documented in Taiwan³⁷ and Hong Kong.³⁸ Despite this growing body of evidence on the association between TV and physical health outcomes, its impact on mental health impact of TV, particularly in subtropical regions, remains poorly understood, highlighting a critical gap in the current literature.

Prior research on the mental health impact of temperature exposure has often evaluated effect modification by demographic characteristics and seasonality,^{3,11,39} revealing differences in vulnerability across subgroups. In the context of Texas, evidence suggests that demographic factors are more strongly associated with vulnerability to temperature-related health incidents compared to other factors.⁴⁰

Additionally, this vulnerability may be amplified by seasonal variations.⁴¹ Specifically, regarding the impact of short-term TV on mental health, existing research has indicated that TV exposure is more strongly associated with mood disorders among younger individuals,^{12,13} and with schizophrenia among men.^{9,10} However, while studies on this topic have frequently stratified by sex and age, potential effect modification by demographic factors and seasonality in subtropical climates remains comparatively underexamined.

Accordingly, to fill these research and knowledge gaps, we conducted a multi-county time-series study in Texas in the United States with the dual aims of 1) evaluating the exposure-lag-response relationship between intra-day temperature variability and outpatient hospital visits for mental and behavioral disorders in subtropical climates and 2) assessing potential effect modification by demographic characteristics (i.e., sex, age, and race) and season.

2. Method

2.1. Study area

In this study, all Texas counties with population size that consistently exceeded 65,000 from 2015 to 2019 were selected, yielding a total of 53 counties. The study areas are mapped in Supplementary Figure S1.

2.2. Data and variables

2.2.1. Mental health outcomes

Data on outpatient hospital visits were obtained from the Texas Department of State Health Service (DSHS) for the period 2015-2019. The Texas DSHS outpatient data includes both emergency department visits and outpatient clinic visits. This data provides comprehensive statewide coverage, as all hospitals are mandated to report outpatient visits, ensuring completeness of the dataset. We identified mental health-related outpatient visits using both principal diagnosis and up to three additional diagnoses of each individual record, according to the codes from the International Classification of Diseases, 9th and 10th Revisions (ICD-9 and ICD-10). Consequently, a single visit could be classified into multiple diagnostic categories if more than one qualifying diagnosis was recorded during the same healthcare encounter. We

further conducted a sensitivity analysis restricted to principal diagnosis only, which demonstrated that the observed association patterns were generally consistent with those of the main analysis (Supplementary Table S5.5). The outcomes examined included overall mental and behavioral disorders (ICD-10: F00-F99) and three diagnostic categories: substance use disorders (ICD-10: F10-F19), mood disorders (ICD-10: F30-F39), and anxiety and stress-related disorders (ICD-10: F40-F49). These categories were selected because they occurred more frequently during the study period, each with over one million recorded visits (Table 1). The specific ICD-9 codes used to identify mental health outcomes are listed in Supplementary Table S1.

We also retrieved data on patient age, sex, and race. After aggregating the outpatient hospital visit records into daily counts, treating repeat visits as separate instances, we further categorized the records by age group (0-19, 20-64, ≥ 65 years), sex group (male and female), and race group (White and non-White), and calculated the corresponding counts for each county. Records with missing or unknown values for age, sex, or race were excluded from the analysis. This study was approved by the Texas DSHS Institutional Review Board (IRB No. 23-009).

2.2.2. Temperature variability exposure metrics

We measured intra-day temperature variability using two different intra-day TV exposure metrics:¹² 1) hourly temperature variability (HTV), which represents intra-day temperature fluctuations and is calculated as the standard deviation of hourly measurements within the exposure day, and 2) diurnal temperature range (DTR), defined as the difference between the daily maximum and minimum on the exposure day. Given the observed high correlation of HTV and DTR (Supplementary Figure S2) and the ability of HTV to capture temperature fluctuations at a finer temporal resolution,^{27,42} we selected HTV as the primary exposure measurement. DTR was included in sensitivity analyses to compare effect estimates and evaluate the robustness of our findings.

Temperature data was obtained from the National Aeronautics and Space Administration (NASA), which provides the primary forcing hourly data of the North American Land Data Assimilation System

(NLDAS-2).⁴³ Temperature data were in 1/8th degree grid spacing (12-14 km) and hourly temporal resolution. We further aggregated these hourly measurements to the county level by averaging the values from all grids within each county. We utilized unweighted weather metrics rather than population-weighted measurement, as the literature reported that temperature-health associations were not greatly different when estimated using unweighted versus population-weighted temperature metrics at the county or regional level.^{44,45}

2.2.3. Covariates

Existing studies have suggested increases in mental distresses to be associated with a number of meteorological parameters, such as air temperature, relative humidity and wind speed.⁴ Environmental factors such as extreme weather events and air pollutants have likewise been linked to poor mental health.⁴⁶ To account for the potential confounding effects of environmental factors, we calculated the daily mean temperature, relative humidity, wind speed, and daily mean concentrations of fine particulate matter (PM_{2.5}), NO₂, and SO₂. We also included binary variables for the presence of weather-related hazards on each exposure day, considering both heat-related hazards (e.g., heat, excessive heat) and cold-related hazards (e.g., cold/wind chill, heavy snow). In addition, prior research has consistently emphasized a strong relationship of socioeconomic status (SES) with mental health disorders.⁴⁷ For example, low socioeconomic status characterized by low income and unemployment rate is associated with increased risks of mental health conditions.⁴⁸ Existing evidence further indicates that the impact of temperature on mental health is heterogeneous across educational attainment, household income, health insurance coverage, and other factors related to healthcare access.⁴⁹⁻⁵² Moreover, mental health impacts of temperature are substantially greater among individuals with disabilities compared with non-disabled populations.⁵³ Therefore, we adjusted for the potential influence of SES and selected indicators based on previous studies,⁵⁴ including poverty level, unemployment status, high-school education attainment, insurance coverage, civilian with a disability, and vehicle ownership. Data sources for covariates are presented in Supplementary Table S2.

2.3. Statistical analysis

A time-series meta-analytic approach was employed for multi-county analyses of the association between daily TV and mental health outcomes. This analytic framework has been widely adapted in previous multi-region studies.^{27,42,55,56}

2.3.1. County-specific models

To develop county-specific models, a two-step process was followed. First, a generalized additive model (GAM) with a log link and spline function for TV was fitted separately for each county to examine and visualize the exposure-response relationship. Based on the results of these initial analyses (Supplementary Figure S3.1-S3.4), a linear function was finally adopted to model the association between TV and mental health outcomes. To further account for potential delayed and distributed effects, we applied GAMs with distributed lag non-linear models (DLNMs), which enable the flexible modeling of exposure-lag-response relationships.⁵⁷ The county-specific model, based on the quasi-Poisson regression and adjusted for potential confounders, is specified as follows:

$$\begin{aligned} \log[E(Y_t)] = & \alpha + cb(HTV_{t,l}) + cb(Tmean_{t,l}) + ns(RH_t, 3) + ns(WS_t, 3) + ns(SO_{2t}, 5) + ns(NO_{2t}, 5) + \\ & ns(PM_{2.5t}, 5) + ns(time_t, 7 \times year) + \delta Hazards_t + \varepsilon SES_t + \epsilon DOW_t + \theta Holiday_t \end{aligned} \quad (1)$$

where $E(Y_t)$ refers to the daily count of outpatient hospital visits on day t ; $cb()$ is a cross-basis function; $ns()$ denotes the natural cubic spline (NS); $HTV_{t,l}$ represents the daily TV on day t and lag day l , wherein a NS with 5 degrees of freedom (df) over 7 lag days was applied to capture potential delayed effects, as previous studies have reported lag effects of TV exposure extending up to seven days in relation to risk of mental health disorders;^{12,13} $Tmean_{t,l}$ represents the daily mean temperature on day t and lag l , modeled using a NS with three internal knots placed at the 10th, 75th, and 90th percentiles, and a NS with 3 df for a 21-day lag period; a succession of NS terms adjusting for environmental factors: one with 3 df to adjust for daily relative humidity and wind speed, two with 5 df to control for respective air pollutant concentrations, and one with 7 df per year to consider seasonal and long-term trends; $Hazards$ represents a categorical variable accounting for occurrence of a weather-related hazard warning on day t ; SES

denotes the percentage of the population: below poverty level, unemployed, without a high school diploma, without health insurance, with a disability, and without a vehicle; *DOW* denotes a categorical variable referring to the day of the week; and *Holiday*, a binary variable indicating the declines in hospital visits during holidays.

We further performed sensitivity analyses to test the robustness of the results by using a modified 28-day lag structure for daily mean temperature and varying df for relative humidity, wind speed, air pollutant concentrations, and time. If the association between HTV and mental health outcomes continued to increase over the initial 7-day lag period, we also extended the HTV lag window to 14 days to further evaluate the lag-response relationship.

Multicollinearity among covariates was assessed using variance inflation factors (VIFs). No substantial multicollinearity was detected, as all VIF values were below 4 (Supplementary Table S7).⁵⁸

Following analytic approaches used in the literature for subgroup analyses,^{12,59,60} we explored the potential effect modification of demographic characteristics via stratifying analyses by sex, age, and race, and seasonal variation by stratifying analyses according to season: spring (March to May), summer (June to August), fall (September to November), and winter (December to February). These subgroup analyses were conducted on an exploratory basis to assess potential heterogeneity in the associations between HTV and mental health-related outpatient hospital visits.

2.3.2. Meta-analysis

County-specific effect estimates were pooled to derive state-level effect estimates through meta-analysis, specifically a random-effects meta-analysis using the restricted maximum likelihood method. We conducted separate meta-analyses for each type of mental disorder across different exposure days (lag 0 to lag 7). We also pooled county-specific effect estimates from stratified analyses to evaluate subgroup-specific associations at the state level. Final results were reported as percentage change in mental disorders per 1 °C increase in HTV, along with 95% confidence intervals (95% CIs).

2.3.3. Difference test on demographic and seasonal characteristics

Subgroup differences were assessed using Wald tests within random-effects meta-regression models, testing the null hypothesis that pooled associations were equal across subgroups. For comparisons involving two groups (e.g., sex, race), Wald z-tests were applied to evaluate pairwise differences in pooled effect estimates. For comparisons involving more than two groups (e.g., age, season), Wald χ^2 test was conducted to assess overall heterogeneity across subgroups. This method has been widely used in meta-analytic studies examining effect modification in the association between temperature exposure and health outcomes.⁶¹⁻⁶⁴

All statistical analyses were performed in R Studio (version 4.2.0). We used the *mgcv* package for GAMs, *dlm* for DLNMs, *metafor* for meta-analyses, and *ggplot2* for data visualization.

3. Results

3.1. Descriptive statistics

For all mental disorders and three specific mental disorder categories, most outpatient hospital visits during 2015-2019 in the 53 Texas counties occurred among adults aged 20-64 years. Overall, a slightly higher proportion of visits for overall MBDs were made by females and non-White populations. Substance use disorders were more common among males, while mood disorders and anxiety- and stress-related disorders were more prevalent among females. Hospital visits for mood disorders were also proportionally higher among White populations. Seasonal distributions were relatively balanced (Table 1). During the study period, the annual average HTV was 3.83 °C (range: 0.10-10.37 °C) and the average DTR was 11.34 °C (range: 0.31-31.09 °C). Substantial variability was also observed in annual average daily mean temperature, relative humidity, wind speed, and air pollutant concentrations. Heat- and cold-related hazard warnings were infrequent, while SES indicators such as poverty rate and insurance coverage showed notable disparities across counties (Supplementary Table S3).

[Table 1 is about here.]

3.2. Association of temperature variability with mental and behavioral disorders

Figure 1 presents the state-level percentage changes in outpatient hospital visits for mental and behavioral disorders associated with 1 °C increase in HTV at different exposure lag days. The effect estimates in each county can be found in Supplementary Table S4.1-S4.4. The highest cumulative risks were observed over lag 0-1 days for overall MBDs and for the three diagnostic categories, indicating an acute and immediate impact of HTV exposure.

For overall MBDs, a 1 °C increase in HTV was associated with a cumulative 0.71% increase in outpatient hospital visits (95% CI 0.58-0.85) over lag 0-1 days. This cumulative effect peaked within two days and then declined monotonically across subsequent lags. Statistically significant associations persisted over lag 0-6 days, after which the estimates dissipated and became non-significant by lag 0-7 days.

For substance use disorders, a 1 °C increase in HTV was associated with a cumulative 0.63% increase in outpatient hospital visits (95% CI 0.47-0.79) over lag 0-1 days. Similar to the pattern observed for overall MBDs, the effect estimates declined monotonically across subsequent lags and remained statistically significant over lag 0-5 days.

For mood disorders, a 1 °C increase in HTV was associated with a cumulative 0.67% increase in outpatient hospital visits (95% CI 0.35-0.99) over lag 0-1 days. The effect estimate declined sharply after lag 0-2 days and was no longer statistically significant beyond this period.

For anxiety- and stress-related disorders, a 1 °C increase in HTV was associated with a cumulative 0.74% increase in outpatient hospital visits (95% CI 0.48-1.01) over lag 0-1 days. Consistent with the patterns observed for overall MBDs and substance use disorders, the cumulative effect peaked within two days and declined monotonically across subsequent lag periods. Statistically significant associations persisted through lag 0-3 days.

[Figure 1 is about here.]

3.3. Effect modification by demographic characteristics

Results of the state-level stratified analyses by demographic characteristics are presented for lag 0-1 days, the exposure period associated with the highest cumulative risk. Detailed between-subgroup comparisons by demographic characteristics are provided in Supplementary Table S6.1- S6.3.

Sex differences. Figure 2 presents the effect modification by sex. Across all outcomes, a 1 °C increase in HTV was significantly associated with elevated risks for outpatient hospital visits among both males and females. However, no statistically significant differences were observed between the two groups for overall MBDs (Wald z-test p -value = 0.352), substance use disorders (Wald z-test p -value = 0.157), mood disorders (Wald z-test p -value = 0.902), and anxiety- and stress-related disorders (Wald z-test p -value = 0.768).

[Figure 2 is about here.]

Age differences. Significant effect modification by age was observed for overall MBDs and substance use disorders (all Wald χ^2 test p -values < 0.05). The impact of a 1 °C increase in HTV on outpatient hospital visits for overall MBDs was greater among children and youths (aged 0-19 years), with a cumulative 1.21% increase in hospital visits (95% CI 0.84-1.58), compared to adults aged 20-64 (0.63%, 95% CI 0.47, 0.79) and older adults (≥ 65 years) (0.92%, 95% CI 0.57-1.28). Particularly, the difference between children/youth and adults aged 20-64 was statistically significant. Stronger associations for substance use disorder-related outpatient visits were observed among older adults, with hospital visits increasing by 1.50% (95% CI: 0.93-2.08), a significantly greater effect than that observed among adults aged 20-64. In contrast, children and youths appeared more vulnerable to outpatient visits for anxiety- and stress-related disorders (1.33%, 95% CI 0.60-2.07), exhibiting the highest effect estimate across age groups. However, no evidence of age-specific differences was observed for mood disorders (Wald χ^2 test p -value = 0.195) and anxiety- and stress-related disorders (Wald χ^2 test p -value = 0.126).

[Figure 3 is about here.]

Race differences. For overall MBDs, substance use disorders, and anxiety- and stress-related disorders, a 1 °C increase in HTV was associated with significantly higher risk of outpatient hospital visits in both White and non-White populations. For mood disorders, a significant association was observed only among the White population. Overall, we observed no statistically significant difference by race for overall MBDs (Wald z-test p -value = 0.966), substance use disorders (Wald z-test p -value = 0.656), mood disorders (Wald z-test p -value = 0.323), and anxiety- and stress-related disorders (Wald z-test p -value = 0.868).

[Figure 4 is about here.]

3.4. Effect modification by season

In addition to demographic parameters, the association between mental health-related outpatient hospital visits and HTV was also found to differ across the four seasons. The effect estimates varied across both lag exposure days and season. The highest cumulative percentage change in outpatient hospital visits was observed at lag 0-1 during in most seasons. For example, in spring and summer, the highest cumulative percentage changes were consistently observed at lag 0-1, whereas in fall, the effects were more acute, with peak estimates occurring at lag 0 for all outcomes except substance use disorders, which peaked at lag 0-1. In winter, the cumulative risks increased progressively through lag 0-7 days, suggesting more delayed and prolonged effects during colder months. Detailed association of HTV with outpatient hospital visits for overall MBDs and the three diagnostic categories across different exposure lag days and seasons are presented in Supplementary Figure S4. Figure 5 presents the results of the state-level stratified analyses by season over lag 0-1 days. Detailed pairwise comparisons between seasonal groups are presented in Supplementary Table S6.4.

Seasonal differences. Significant effect modification by season was observed for overall MBDs, mood disorders, and anxiety and stress-related disorders (all Wald χ^2 test p -values < 0.05). For overall MBDs, a 1 °C increase in HTV was associated with higher risk of outpatient hospital visits in spring (0.52%, 95%

CI 0.32-0.71) and summer (0.45, 95% CI 0.18-0.71) compared with other seasons. A statistically significant contrast was observed among the two transition seasons, with the cumulative risk in spring higher than in fall. For substance use disorders, a similar seasonal pattern was found, with stronger associations for a 1 °C increase in HTV during spring (0.45, 95% CI 0.21-0.69) and summer (0.42%, 95% CI 0.10-0.75). However, the impact of HTV on outpatient hospital visits for mood disorders and anxiety- and stress-related disorders was lower in summer and relatively higher in both spring and winter, with no statistically significant association observed in fall. For mood disorders, a 1 °C increase in HTV was associated with a cumulative increase of outpatient hospital visits, at 0.64% (95% CI 0.25-1.03) in spring and 0.59% (95% CI 0.24-0.95) in winter, both significantly higher than estimate effects observed in fall. For anxiety- and stress-related disorders, effect estimates were 0.68% (95% CI 0.36-0.99) in spring and 0.54% (95% CI 0.13-0.94) in winter. A statistically significant difference between spring and fall was also observed for anxiety disorders.

[Figure 5 is about here.]

3.5. Sensitivity analysis results

Our results were robust to a modified 28-day lag structure for daily mean temperature and to changes in the df for relative humidity, wind speed, air pollutant concentrations, and time. When TV exposure was measured as the interquartile range increase in temperature variability, the overall patterns remained consistent at the state level (Supplementary Table S5.1-S5.2). Replacing HTV with DTR in the model also yielded consistent patterns; for overall MBDs and the three diagnostic categories, the highest risk in association with a 1 °C increase in DTR occurred across lag 0-1 days (Supplementary Table S5.3). Additionally, excluding SES variables from the model resulted in effect estimates of similar magnitude and direction, with no changes in statistical significance (Supplementary Table S5.4), indicating that the observed associations were robust to SES adjustment. As the estimates in winter continued to increase through lag 0-7 days, we also examined the 14-day lag structure for HTV across four seasons. The overall association patterns remained consistent in summer, spring, and fall. In winter, cumulative risks continued

to increase and peaked at lag 0-12 days across all outcomes, confirming more prolonged mental health risks during winter seasons (Supplementary Figure S5).

4. Discussion

This study examined the exposure-lag-response relationships between intra-day TV and mental health-related outpatient hospital visits in Texas. Our findings indicated lag effects of intra-day HTV exposure, with the highest cumulative impacts on all mental health outcomes observed over exposure day and the following day (lag 0-1 days); specifically, a 1 °C increase in HTV was associated with a 0.71% cumulative increase in outpatient hospital visits for overall MBDs during that period. This highlights the acute and immediate effects of TV on mental health. In addition, our analyses of specific mental disorders showed the impacts of HTV on outpatient hospital visits for substance use disorders, mood disorders, and stress-related disorders also increased significantly for lag 0-1 days, with the strongest impact observed for anxiety- and stress-related disorders. Finally, we examined the effect modification by sex, age, race, and season in the association of intra-day TV with mental health-related outpatient hospital visits and identified significant differences across age and seasonal groups.

Our findings are generally consistent with existing studies that reported positive associations of intraday TV with hospital visits for mood disorders and for anxiety-related disorders.^{12,13} These observed associations could be supported by several underlying mechanisms. First, unstable temperature conditions may affect the circadian body temperature rhythm, which plays important roles in mood regulation, cognitive performance, psychological well-being.⁶⁵ Second, temperature fluctuations may stimulate the hypothalamic-pituitary-adrenal axis, leading to elevated cortisol levels and heightened physiological stress,⁶⁶ which are associated with anxiety, depressive symptoms, and impaired emotional regulation.⁶⁷ Third, individuals with pre-existing mental health conditions often appeared particularly vulnerable, as some psychiatric drugs (e.g., antipsychotics, antidepressants) may impair hypothalamic thermoregulation, thus compromising the body's ability to maintain homeostasis and increasing the risk of hospitalization.⁶⁸

We also observed age-related differences in the association between intra-day HTV exposures and outpatient visits for mental disorders. Susceptibility to different types of mental health disorders likewise varies across age; for example, older adults were more susceptible to HTV-related outpatient hospital visits for substance use disorders compared with adults aged 20-64 years. Although prior research has reported associations of increased daily temperature and higher risk of substance use disorder-related emergency healthcare service utilization, subgroup heterogeneity has been largely unexplored in the existing literature.^{4,69} Additionally, to our knowledge, no studies to date have specifically focused on substance use disorders in relation to temperature variability, limiting evidence to support our findings in this area. Therefore, our findings would extend the current literature and underscore potential age-specific vulnerabilities. We further observed that children and youth were at greater risk of outpatient hospital visits for overall MBDs associated with HTV than adults aged 20-64 years. Consistent with previous findings from studies conducted in China and New York state,^{12,13} stronger associations between TV and mental disorders have often been reported among younger groups.

Our stratified analysis by season additionally identified that more immediate effects occurred on the same day or by the following day (lag 0 or lag 0-1) during spring, summer, and fall, with spring generally showing higher risks of outpatient hospital visits for mental disorders associated with HTV. This seasonal difference may be due to spring featuring overall more abrupt temperature changes and rapid increases in daylight, which destabilize circadian rhythms and melatonin secretion.⁷⁰ During winter months, more delayed effects were observed, with the greatest cumulative risk appearing over 0-12 days. We additionally observed a clear contrast of the two transition seasons, spring and fall. For overall MBDs and substance use disorders, HTV-related risks were notably higher in spring than in fall; for mood disorders and anxiety disorders, risk estimates likewise remained elevated in spring. Prior research on seasonal modification of the effect of TV has often grouped spring and fall together as a collective transition/moderate season, reporting associations with health outcomes during these periods in comparison to summer/hot and winter/cold seasons.^{12,55,71} However, our findings indicate substantial

heterogeneity between spring and fall, suggesting that future research should consider and analyze these two transition seasons separately to produce more precise season-specific estimates and support more reliable adaptation planning.

Several limitations of this study should be acknowledged. First, the aim was to capture intraday variations in temperature, and we therefore used weather data from NLDAS-2, which provides hourly temporal resolution but a relatively coarse spatial resolution. Although this approach is standard for county level time-series analysis, it may not fully capture within-county spatial heterogeneity in temperature exposure, particularly in large Texas counties with substantial geographic and population variations. As a result, the use of county-level average metrics rather than population-weighted exposure estimates may overlook finer-scale climate variations within counties and introduce exposure misclassification. Second, relying on outpatient hospital visits as the outcome measurement in this study may underestimate the full mental health burden of temperature variations, as individuals may not seek medical care or face barriers such as facility access, lack of insurance, stigma, or underdiagnosis. Third, as temperature variability differs substantially across geographical regions,⁵⁵ associations observed in Texas may not be generalizable to other climates and populations.

In conclusion, this study highlights the burden of mental disorders associated with temperature variability in Texas. In particular, our findings demonstrated greater susceptibility of youth and older adults to TV-related outpatient hospital visits for MBDs, emphasizing the need for tailored public health interventions to address mental health disparities. The seasonal differences observed further underscore the need to pay particular attention to the heightened risks in spring and prolonged risks in winter. Specifically, given the elevated risks observed during spring, public health agencies should consider integrating mental health-related risk alerts into existing meteorological early warning systems and strengthening crisis service capacity during periods of high temperature variability. In light of the delayed winter effects, primary care providers and community health services may consider enhancing mental health screening and outreach, particularly among vulnerable population such as older adults, in the short-term period following a cold

spell or high temperature variability events. In addition, considering the heightened vulnerability observed among children and adolescents, school-based mental health services could incorporate seasonal monitoring protocols during high-risk periods. Educational institutions and pediatric providers may also strengthen screening and intensify early intervention efforts following periods of pronounced temperature variability. Targeted and adaptive prevention strategies will be critical to mitigate the mental health burden associated with increasing climate variability in subtropical regions.

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Table 1. Summary of demographic characteristics for mental health-related outpatient hospital visits in 53 Texas counties for 2015-2019.

Characteristics	All mental and behavioral disorders (n = 6,867,782)	Substance use disorders (n = 4,330,855)	Mood disorders (n = 1,163,574)	Anxiety- and stress-related disorders (n = 1,473,491)
Age group, years				
0-19	679,515 (9.9%)	192,322 (4.4%)	138,654 (11.9%)	174,686 (11.9%)
20-64	5,507,528 (80.2%)	3,893,338 (89.9%)	911,838 (78.4%)	1,142,458 (77.5%)
65+	680,733 (9.9%)	245,192 (5.7%)	113,081 (9.7%)	156,346 (10.6%)
Sex				
Male	3,306,463 (48.1%)	2,358,113 (54.4%)	421,398 (36.2%)	500,398 (34.0%)
Female	3,560,813 (51.8%)	1,972,385 (45.5%)	742,126 (63.8%)	973,036 (66.0%)
Race group				
White	3,219,026 (46.9%)	2,012,564 (46.6%)	601,108 (51.7%)	704,107 (47.8%)
Non-White	3,283,830 (47.8%)	2,103,829 (48.6%)	502,214 (43.2%)	683,560 (46.4%)
Season				
Spring	1,741,271 (25.4%)	1,097,841 (25.3%)	295,272 (25.4%)	375,968 (25.5%)
Summer	1,769,962 (25.8%)	1,134,038 (26.2%)	292,872 (25.2%)	384,958 (26.1%)
Fall	1,719,253 (25.0%)	1,068,652 (24.7%)	298,907 (25.7%)	368,127 (25.0%)
Winter	1,637,296 (23.8%)	1,030,324 (23.8%)	276,523 (23.8%)	344,438 (23.4%)

Figure 1. Pooled estimates from meta-analyses showing the cumulative percent change (95% CIs) in outpatient hospital visits associated with a 1 °C increase in HTV over different exposure lag days in 53 Texas counties during 2015-2019.

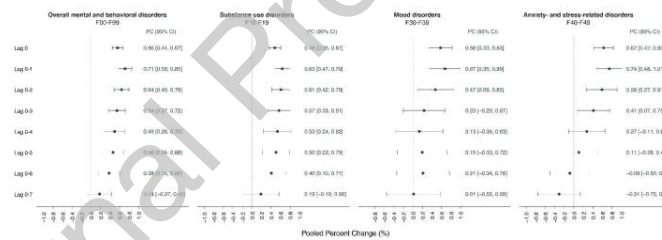


Figure 2. Pooled estimates from meta-analyses showing the cumulative percent change (with 95% CI) in outpatient hospital visits associated with a 1 °C increase over lag 0-1 days, stratified by sex, in 53 Texas counties during 2015-2019.

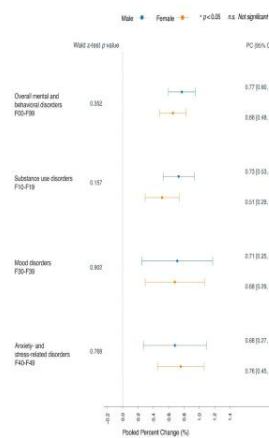


Figure 3. Pooled estimates from meta-analyses showing the cumulative percent change (with 95% CI) in outpatient hospital visits associated with a 1 °C increase over lag 0-1 days, stratified by age, in 53 Texas counties during 2015-2019.

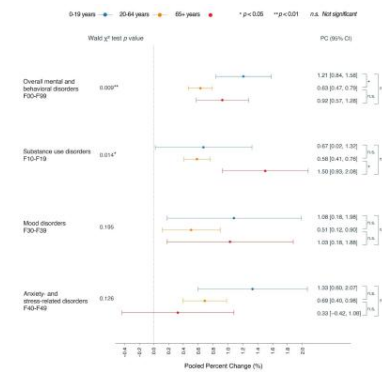


Figure 4. Pooled estimates from meta-analyses showing the cumulative percent change (with 95% CI) in outpatient hospital visits associated with a 1 °C increase over lag 0-1 days, stratified by race, in 53 Texas counties during 2015-2019.

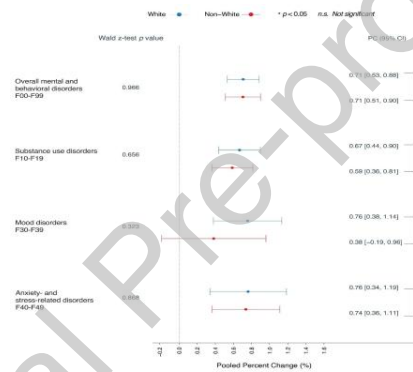
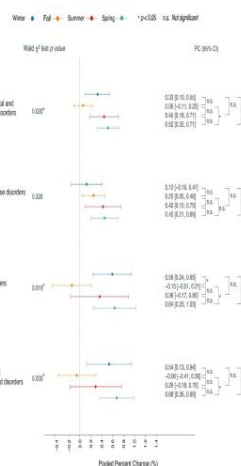


Figure 5. Pooled estimates from meta-analyses showing the cumulative percent change (with 95% CI) in outpatient hospital visits associated with a 1 °C increase over lag 0-1 days, stratified by season, in 53 Texas counties during 2015-2019.



Declaration of Interest Statement

The authors declare no conflict of interest.

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